

CSE/PC/B/T/313: Operating System

Syllabus:

1. Introduction to Operating Systems (including concepts of kernel) [1L] ^[L]_[SEP]
2. Concept of batch-processing, multi-programming, time sharing, real time ^[L]_[SEP] operations [2L] ^[L]_[SEP]
3. Process Management: Concept of process, state diagram, process control block; ^[L]_[SEP] scheduling of processes – criteria, types of scheduling, non-preemptive and preemptive scheduling algorithms like: FCFS, Shortest Job First/Next (SJF/N), Shortest Remaining Time Next (SRTN), Round Robin (RR), Highest Response ratio Next (HRN), Priority based scheduling, different Multilevel queue scheduling etc.; [5L] ^[L]_[SEP]
4. Threads – concept, process vs thread, kernel and user threads, multithreading models [2L] ^[L]_[SEP]
5. Inter-process Communication (IPC) – Shared memory, message, FIFO, concept of semaphore, critical region, monitor [4L] ^[L]_[SEP]
6. Process Synchronization: concepts, race condition, critical section problem and its solutions; synchronization tools- semaphore, monitor etc., discussion of synchronization problems like producer-consumer, readers-writers, dining philosophers, sleeping-barber etc. ^[L]_[SEP] Deadlock – conditions, resource allocation graph, prevention techniques, ^[L]_[SEP] avoidance technique – Banker's algorithm and related algorithms [6L] ^[L]_[SEP]
7. Memory management: Address space and address translation; ^[L]_[SEP] static partitioning, dynamic partitioning, different types of fragmentation, paging, segmentation, swapping, virtual memory, demand paging, page size, page table, page replacement algorithms – FIFO, LRU, Optimal page replacement, Variants of LRU, etc; thrashing, working set strategy [6L] ^[L]_[SEP]
8. File Management: File and operations on it, file organization and access; file allocation; directory structures, file sharing, file protection [4L] ^[L]_[SEP]
9. Device management: Magnetic disks, ^[L]_[SEP] disk scheduling- criteria, algorithms – FCFS, SSTF, SCAN, C-SCAN, LOOK, etc, disk management – formatting, boot block, disk free space management techniques, concept of RAID etc [3L] ^[L]_[SEP]
10. Protection and Security: Concepts of domain, Access matrix and its implementation, access control, Security of systems- concepts, threats- Trojan horse, virus, worms etc, introduction to cryptography as security tool, user authentication [5L] ^[L]_[SEP]

11. Case Study [2L] ^[L]_[SEP]

Suggested Readings:

1. Operating Systems Concepts – A. Silberschatz, P. Galvin and G. Gagne. Wiley India
2. Operating Systems Concepts - Gary Nutt, N. Chaki and S. Neogy, Pearson Education ^[L]_[SEP]
3. Operating Systems – W. Stallings, Pearson Education ^[L]_[SEP]
4. Operating Systems: A Concept-based Approach – D. M. Dhamdhare, Tata ^[L]_[SEP] McGraw-Hill ^[L]_[SEP]

Course Outcomes (COs):

At the end of this course, each student should be able to:

CO1: Comprehend the concept and functionalities of OS. [K2]

CO2: Analyse the issues and discuss challenges of managing the computing resources. [K4]

CO3: Study and explain the policies and compare algorithms for managing resources and demonstrate competence related to scheduling, synchronization, multithreading, memory, disk, file. [K3]

CO4: Identify protection and security issues and corresponding techniques. [K2]

CO5: Explain OS architecture (Linux/Unix/Mach OS/Windows). [K5]

CSE/PC/B/T/313: Operating System: Course Articulation Matrix

OS CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO3
CO1	3	1				1		1	2	2			3		
CO2		3	1										3	3	2
CO3	1	2	3		1								1		
CO4		2		3									2	3	
CO5	2				3				2	2		2	2		1

PSO – 1

With a strong foundation in Computer Science and Engineering, students will have the capability of designing and developing software and hardware systems.

PSO – 2

With exposure to various computational and analytical methods, students will develop the capability of research and innovation.

PSO – 3

Students will be able to address the diverse needs of society with responsibility and professional ethics.